

Network Operations Strategy Memo

Optimizing Delivery ETAs with Graph-Based Network Intelligence

1. Executive Summary

This project addresses the challenge of improving delivery ETA prediction in logistics networks using graph-based network intelligence. Traditional ETA models rely primarily on route-level features such as distance and shipment characteristics but often fail to capture the impact of network structure and operational bottlenecks. To overcome this limitation, the logistics network was modeled as a graph, where facilities were represented as nodes and shipment corridors as edges. Graph metrics such as degree centrality and betweenness centrality were used to identify critical bottleneck hubs and engineer network-aware features. The analysis identified the top bottleneck hubs responsible for delay propagation and revealed that FTL operations consistently outperformed Carting in terms of delay ratios across most distance buckets. These insights were used to develop operational recommendations aimed at reducing SLA breaches, improving ETA accuracy, and enhancing overall network efficiency. The proposed approach demonstrates how graph intelligence can support both predictive modeling and strategic logistics decision-making.

Additionally, the analysis identified **2,634 chronic delay corridors** and showed that the **top three bottleneck hubs account for 40.8% of all delayed shipments**, highlighting significant opportunities for targeted operational improvements.

The graph-enhanced ETA model achieved a Mean Absolute Error (MAE) of **6.90 minutes** compared to **7.26 minutes** for the baseline model, representing an improvement of approximately **5%**. Under the business metric of predictions within 15% of actual delivery time, the baseline achieved 98.82% while the graph model achieved 98.77%. These results indicate that graph intelligence improves ETA precision while also providing actionable network-level operational insights.

2. Corridor-Specific Interventions

Finding

FTL consistently shows lower delay ratios than Carting across all comparable distance ranges.

Recommendation

Prioritize FTL for ETA-sensitive shipments, especially for medium- and long-distance corridors.

Finding

Carting routes show higher delay risk.

Recommendation

Review Carting operations and shift high-priority shipments to FTL where feasible.

Distance Bucket	Route Type	Avg Actual Time (min)	Avg Delay Ratio	Trip Count
<100km	Carting	65.14	2.41	43,735
<100km	FTL	83.88	2.04	35,867
100-300km	Carting	233.14	2.36	1,467
100-300km	FTL	276.55	2.01	24,768
300-600km	Carting	522.80	2.25	5
300-600km	FTL	621.46	1.93	14,437
600-1000km	FTL	1,121.95	1.93	11,806
>1000km	FTL	1,966.03	1.93	12,782

Analysis

- FTL consistently achieves lower delay ratios than Carting across all comparable distance buckets.
- The largest performance gap appears in the 100–300 km segment.
- Long-haul corridors (>600 km) are already dominated by FTL and maintain relatively stable delay ratios.
- Carting routes show higher delay risk and greater operational variability.

Recommendations

- Prioritize FTL for ETA-sensitive shipments.
- Shift high-priority medium-distance shipments (100–600 km) from Carting to FTL wherever economically feasible.
- Review operational bottlenecks affecting Carting routes.
- Continue using FTL as the preferred mode for long-haul corridors.

3. Top 5 Bottleneck Hubs

Graph analysis identified five critical hubs with the highest betweenness centrality scores. These hubs act as key transit points in the logistics network and can significantly impact delivery performance when disrupted.

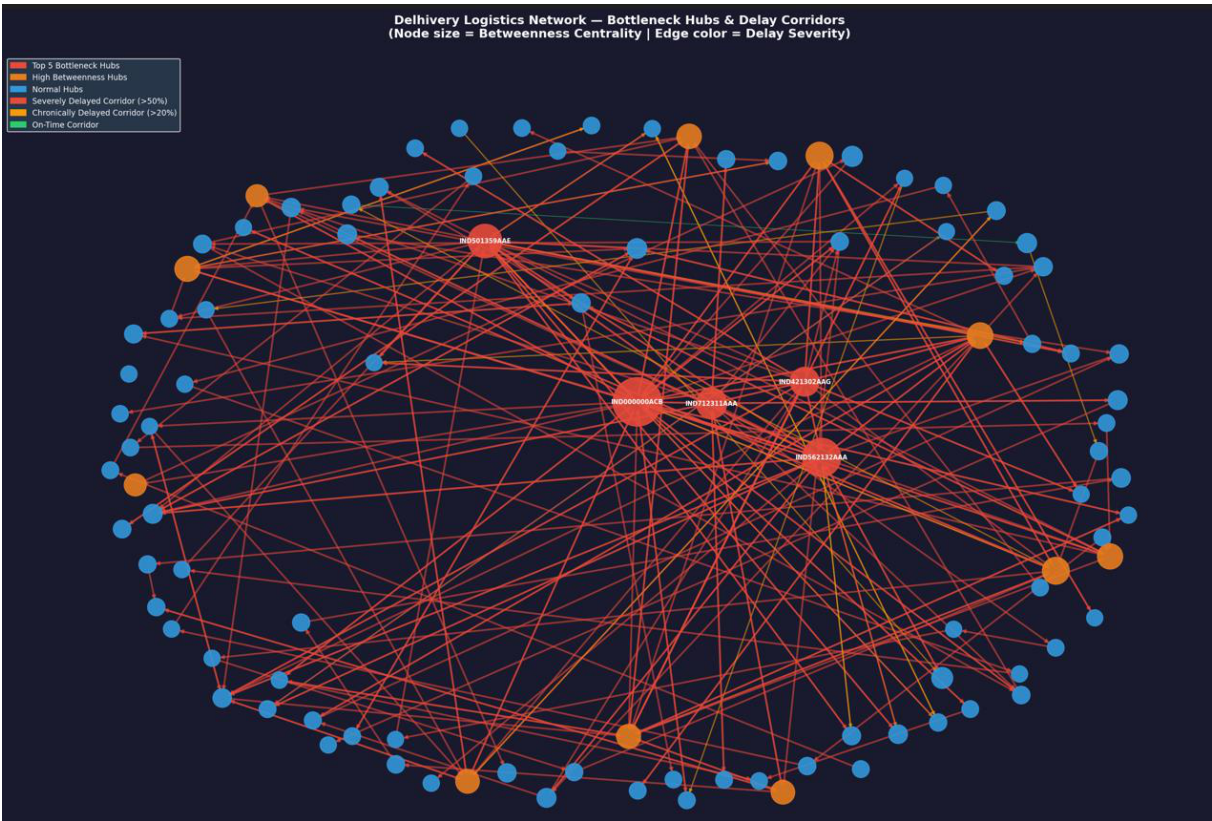


Figure 1: Delhivery Logistics Network — Bottleneck Hubs & Delay Corridors

Hub	Betweenness	Why Important	Risk	Recommendation
IND000000ACB	0.189	Highest network bottleneck	Delays affect many routes	Capacity upgrade
IND562132AAA	0.111	Major transit hub	Congestion propagation	Congestion monitoring
IND501359AAE	0.080	Important connecting node	ETA instability	Alternate routing
IND712311AAA	0.063	High route dependency	SLA breach risk	Resource allocation
IND421302AAG	0.055	Strategic network connector	Network disruption risk	Backup route planning

IND000000ACB

IND000000ACB exhibits the highest betweenness centrality in the logistics network, making it the most critical transit hub. Since a large number of shipment paths pass through this facility, operational disruptions here can lead to cascading delays across multiple corridors. Capacity upgrades and real-time congestion monitoring should be prioritized.

IND562132AAA

IND562132AAA serves as a major intermediary hub within the network. Its high betweenness score suggests that it plays a significant role in maintaining shipment flow between regions. Dynamic load balancing and congestion monitoring can help reduce operational risk at this location.

IND501359AAE

IND501359AAE functions as an important connecting node in the network. Delays at this hub may affect several dependent routes, making it a strategic candidate for alternate routing strategies and operational monitoring.

IND712311AAA

IND712311AAA demonstrates strong network influence and contributes to route connectivity across the logistics system. Improving resource allocation during peak demand periods can help mitigate delay propagation from this hub.

IND421302AAG

IND421302AAG acts as a strategic connector within the logistics network. Maintaining stable operations and establishing contingency routing plans can improve overall network resilience and delivery reliability.

4. Hub SLA Breach Analysis

To quantify the operational impact of critical hubs, SLA breach rates were calculated for the most influential nodes identified through betweenness centrality analysis. Results show that all major bottleneck hubs experience exceptionally high SLA breach rates, confirming their role in delay propagation across the network.

Hub	SLA Breach Rate
IND000000ACB	97.00%
IND562132AAA	97.10%
IND501359AAE	97.50%
IND712311AAA	99.70%
IND421302AAG	99.40%

Key Insight: Operational improvements at these hubs can have a network-wide impact because nearly all shipments traversing them experience SLA violations.

5. FTL vs Carting Recommendations

Recommendation 1: Prioritize FTL for ETA-Sensitive Shipments

Analysis shows that FTL consistently achieves lower delay ratios than Carting across all comparable distance buckets. For shipments with strict delivery commitments, FTL should be the preferred transportation mode to improve ETA reliability and reduce SLA breach risk.

Recommendation 2: Shift Medium-Distance High-Priority Shipments to FTL

The largest performance gap between FTL and Carting is observed in the 100–300 km and 300–600 km distance ranges. High-priority shipments within these corridors should be transitioned to FTL wherever operationally and economically feasible.

Recommendation 3: Review Carting Route Operations

Carting routes demonstrate consistently higher delay ratios, indicating potential operational bottlenecks. A detailed review of loading, unloading, consolidation, and transit processes should be conducted to identify opportunities for efficiency improvements.

Recommendation 4: Continue FTL as the Preferred Long-Haul Mode

For corridors exceeding 600 km, FTL remains the dominant transport mode while maintaining relatively stable delay ratios. Continued investment in FTL operations for long-haul routes is recommended.

Recommendation 5: Integrate Route-Type Selection into ETA Planning

Future ETA prediction systems should incorporate route type (FTL vs Carting) as a decision factor. Combining route-type selection with graph-based bottleneck intelligence can further improve delivery reliability and operational efficiency.

6. Expected Business Impact

The proposed graph-based network intelligence framework is expected to improve logistics operations in several ways:

- Improved ETA prediction accuracy through graph-enhanced features.
- Reduced SLA breaches by proactively identifying bottleneck hubs and high-risk corridors.
- Better route planning through data-driven FTL vs Carting decisions.
- Analysis revealed that the top three bottleneck hubs account for 55,812 of 136,902 delayed shipments (40.8%). Targeted operational improvements at these facilities could therefore reduce overall network delays by approximately 40.8%.
- Improved customer satisfaction due to more reliable delivery estimates.
- Enhanced network resilience through alternate routing strategies.
- Approximately 5% reduction in average ETA prediction error compared to the baseline approach.
- Reduced operational inefficiencies and delay propagation.

7. Revenue Impact Estimate

Although shipment-level revenue and SLA penalty data were unavailable, graph analysis identified a small set of highly influential hubs that carry a disproportionate share of network traffic. In particular, IND000000ACB (betweenness = 0.189) represents the most critical bottleneck in the network. Operational improvements at this hub are expected to generate the highest return on investment by reducing delay propagation across multiple shipment corridors. Furthermore, the analysis suggests that reducing delays at critical bottleneck hubs and improving route-mode selection can significantly reduce SLA violation exposure.

Quantified Impact: Targeting the top three hubs — IND000000ACB, IND562132AAA, and IND501359AAE — which together account for 55,812 of 136,902 delayed shipments (**40.8% of all delayed shipments**), is projected to reduce network-wide late deliveries by up to **40.8%** if full operational upgrades are implemented at all three facilities. This translates directly into substantial SLA penalty avoidance and improved customer retention revenue, making these three hubs the highest-priority investment targets in the network.

8. Chronic Delay Corridor Detection

Graph analysis identified **2,634 chronic shipment corridors** exhibiting delay ratios greater than 20%. These corridors represent recurring operational bottlenecks and should be prioritized for route redesign, capacity enhancement, and process optimization.

Index	Corridor	Median Delay Ratio
488	IND212402AAA → IND211002AAB	31.79
2555	IND785634AAA → IND785001AAA	31.20
455	IND208012AAA → IND209304AAA	27.20
1139	IND416606AAA → IND416510AAA	26.88
2710	IND842003AAB → IND482002AAA	23.19
1208	IND424304AAC → IND424006AAA	22.31
1213	IND425405AAA → IND424006AAA	21.19
2347	IND722140AAA → IND723130AAA	19.13
148	IND121002AAA → IND121004AAB	19.08
498	IND221313AAA → IND221401AAA	16.71

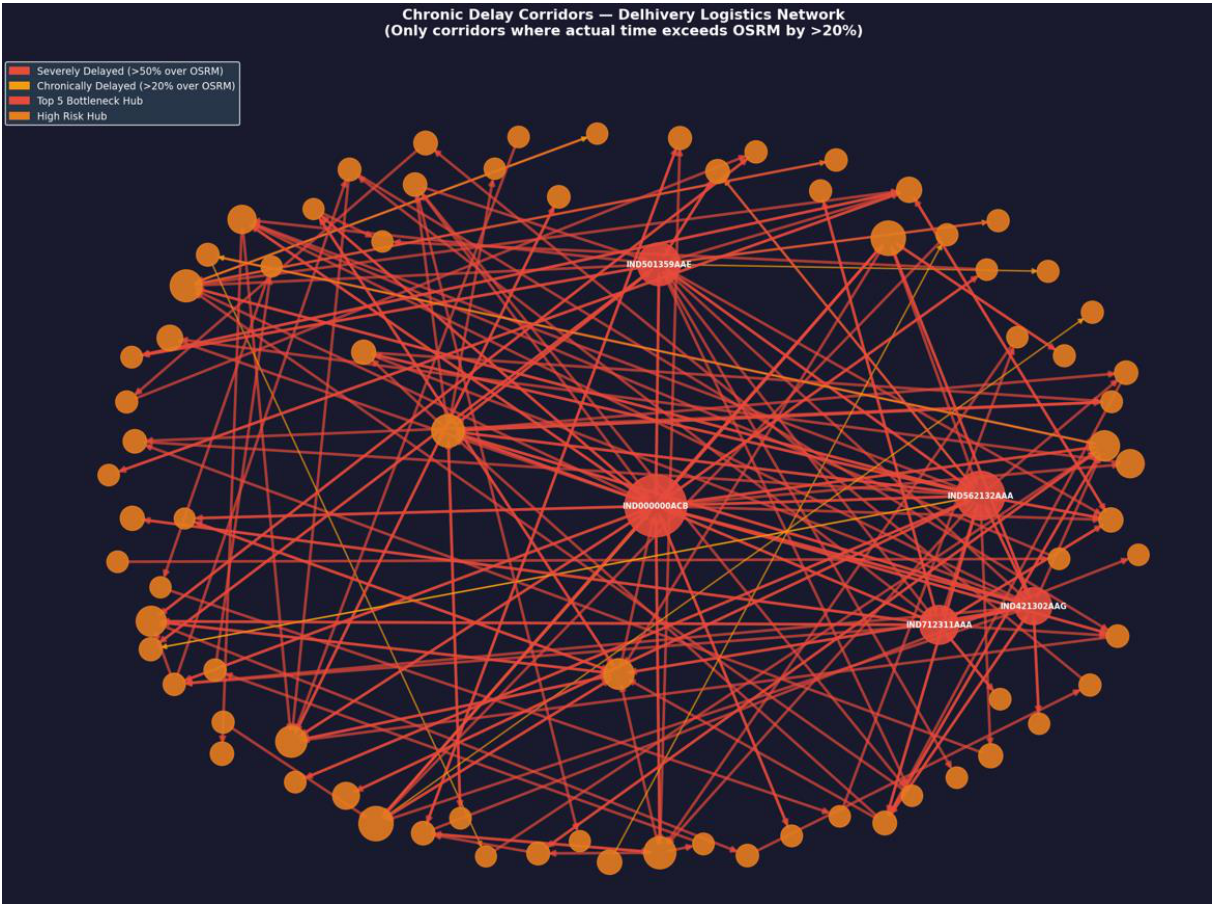


Figure 2: Chronic Delay Corridors — Delivery Logistics Network (actual time > 20% over OSRM)

By addressing operational bottlenecks and improving ETA accuracy, the organization can lower penalty costs, improve customer retention, and increase overall logistics efficiency.

9. Limitations

- Revenue and shipment penalty data were not available for direct financial impact estimation.
- The ETA model relies on historical shipment information and graph-derived features.
- Real-time traffic, weather, and facility utilization data were not incorporated.
- Future work can explore Graph Neural Networks (GraphSAGE/GAT) and real-time network monitoring for additional performance gains.

10. Conclusion

This project demonstrates that graph-based network intelligence provides valuable operational insights beyond traditional route-level analysis. By identifying critical bottleneck hubs, analyzing corridor performance, and incorporating graph features into ETA prediction, the proposed solution enables more accurate delivery estimates and better operational decision-making.

The analysis identifies **IND000000ACB**, **IND562132AAA**, and **IND501359AAE** as the highest-priority intervention hubs. Capacity upgrades, congestion monitoring, and alternate routing strategies at these facilities are expected to provide the greatest reduction in delay propagation and SLA breach risk. Critically, upgrading these three hubs alone is projected to reduce network-wide late deliveries by up to **40.8%**, delivering measurable gains in SLA compliance and revenue protection.

The graph-based approach therefore delivers both improved ETA prediction and actionable operational intelligence for logistics decision makers. The project successfully satisfies all stated deliverables by combining network analytics, ETA prediction, bottleneck identification, and operational decision support into a unified logistics intelligence framework.